

SHIMIZU, Japan

WMO: 478980

Lat: 32.7231N

Lon: 133.0103E

Elev: 111

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	35.1	37.3	17.2	13.2	41.5	19.5	14.7	42.2	22.3	57.6	19.0	55.7	7.3	0	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.2	87.7	79.6	86.4	79.1	85.3	78.6	81.2	85.3	80.5	84.5	79.7	83.8	7.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	156.7	84.0	79.2	152.3	83.5	78.5	148.4	83.0	45.0	85.4	44.2	84.6	43.4	84.0	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.7	17.7	15.3	DB	31.1	91.1	2.2	1.7	29.5	92.3	28.2	93.3	27.0	94.3	25.4	95.5	(4)
(5)			WB	27.0	83.0	2.1	1.5	25.4	84.1	24.2	85.0	23.0	85.8	21.5	86.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	47.8	49.6	54.6	62.0	68.9	73.5	80.2	82.3	78.2	70.7	61.6	52.2		
	DBStd	12.59	5.58	6.25	5.67	4.67	3.00	3.08	2.76	2.13	3.22	4.29	5.76	6.15		
	HDD50	259	109	78	22	1	0	0	0	0	0	0	1	48		
	HDD65	1939	533	432	323	108	5	0	0	0	0	7	133	398		
	CDD50	5813	41	66	165	361	585	706	936	1000	847	643	347	116		
	CDD65	2017	0	0	1	18	125	256	471	535	397	185	29	0		
	CDH74	15349	0	0	0	4	120	912	4525	6007	3222	543	16	0		
	CDH80	3426	0	0	0	0	3	40	968	1838	554	23	0	0		
(14) Wind	WSAvg	7.7	7.6	7.9	8.3	8.0	7.6	7.7	7.4	7.3	7.5	7.6	7.6	7.6		
(15) Precipitation	PrecAvg	97.0	3.6	4.4	6.9	9.2	10.0	13.2	8.9	9.7	13.3	9.2	5.3	3.0		
	PrecMax	144.7	14.3	13.0	16.3	17.8	19.5	23.7	27.4	38.3	42.3	29.6	16.4	7.7		
	PrecMin	59.1	0.3	1.3	1.3	2.4	3.1	5.9	1.4	0.7	2.1	1.6	1.2	0.1		
	PrecStd	19.9	2.9	2.7	3.3	3.7	4.0	4.7	5.4	7.6	7.9	5.8	3.2	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.6	68.1	70.1	74.0	79.0	83.1	89.5	89.9	87.7	82.6	76.4	69.3		
		MCWB	60.7	63.0	62.2	65.3	66.8	77.0	78.7	79.9	79.2	76.0	71.2	63.6		
	2%	DB	62.0	65.4	67.5	72.0	76.6	81.1	87.0	88.2	85.9	80.1	73.4	66.1		
		MCWB	56.6	61.0	61.4	64.8	67.3	76.0	79.1	79.9	77.9	73.3	66.5	60.0		
	5%	DB	58.9	62.8	65.7	70.6	75.2	79.6	85.5	87.1	84.3	78.5	71.8	63.4		
		MCWB	52.5	57.3	59.5	63.9	67.6	74.9	78.8	79.6	77.0	71.9	65.1	56.7		
	10%	DB	56.6	59.6	63.9	69.1	73.9	78.3	84.3	86.0	83.0	77.0	70.0	61.3		
		MCWB	49.9	53.3	57.6	62.5	67.2	73.9	78.3	79.2	76.5	70.1	63.5	54.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.2	64.7	66.0	70.1	73.7	78.4	82.0	82.7	81.4	78.1	72.0	65.8		
		MCDB	64.7	66.7	67.8	71.9	75.4	80.8	85.9	87.1	85.2	80.9	75.2	67.7		
	2%	WB	58.3	61.7	63.3	68.2	72.0	77.3	80.8	81.6	79.9	76.0	68.8	61.4		
		MCDB	61.3	64.8	66.0	70.5	74.0	79.7	84.8	85.9	83.7	79.0	71.9	64.8		
	5%	WB	54.2	58.4	61.0	65.9	70.7	76.4	79.9	80.9	78.9	73.7	66.7	58.2		
		MCDB	57.7	62.3	64.7	69.1	73.2	78.8	83.7	85.0	82.7	77.3	70.6	62.3		
	10%	WB	50.7	54.4	58.7	64.1	69.2	75.2	79.1	80.2	78.1	71.6	64.7	55.5		
		MCDB	55.6	58.6	63.0	67.9	72.4	77.8	82.8	84.3	81.9	75.8	69.0	60.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	11.4	11.4	10.3	9.0	7.0	6.5	7.2	7.8	8.7	9.9	10.7		
		MCDBR	12.8	12.0	11.7	10.9	10.5	7.7	8.1	8.5	8.6	9.2	9.9	11.0		
		MCWBR	12.4	12.5	11.4	9.6	8.6	5.7	5.3	5.5	6.1	7.6	8.9	10.2		
		MCDBR	11.9	11.7	10.5	9.1	8.0	6.2	7.3	7.6	7.7	7.7	8.7	10.4		
	5% WB	MCWBR	13.1	13.1	11.9	9.6	7.9	5.0	5.1	5.1	5.9	7.3	9.1	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.400	0.444	0.500	0.524	0.534	0.547	0.534	0.512	0.485	0.460	0.419	0.390		
	taud		2.196	2.095	1.949	1.946	1.992	2.048	2.123	2.195	2.228	2.200	2.237	2.238		
	Ebn at Noon		252	252	247	247	246	241	244	248	249	245	244	247		
	Edh at Noon		37	45	57	59	57	54	50	46	43	41	36	34		
(44) All-Sky Solar Radiation	RadAvg		808	1000	1325	1552	1629	1365	1731	1757	1393	1098	847	732		
	RadStd		70	84	100	159	170	197	198	140	136	117	83	53		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+1.55	N/A	+0.70	+0.85	N/A	N/A	N/A	N/A
Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page